

FAULT CODE 2185 - Sensor Supply 4 Circuit - Voltage Above Normal or Shorted to High Source

Warnings and Cautions

CAUTION

To reduce the possibility of damaging a new ECM, all other active fault codes must be investigated prior to replacing the ECM.

CAUTION

To reduce the possibility of pin and harness damage, use the following test leads when taking a measurement: Part Number 3164133 - male Deutsch™/AMP™/Metri-Pack™ test lead, Part Number 3822758 - male Deutsch™/AMP™/Metri-Pack™ test lead, and Part Number 3822917 - female male Deutsch™/AMP™/Metri-Pack™ test lead.

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Check the fault codes.	
	STEP 1A. Check for an active fault code.	Fault Code 2185 active?
STEP 2.	Check the ECM and engine harness.	
	STEP 2A. Inspect the ECM and engine harness connector pins.	Dirty or damaged pins?
	STEP 2B. Check the ECM response.	Fault Code 2185 active?
STEP 3.	Clear the fault codes.	
	STEP 3A. Disable the fault code.	Fault Code 2185 inactive?

STEPS		SPECIFICATIONS
	STEP 3B. Clear the inactive fault codes.	All fault codes cleared?

STEP 1. Check the fault codes.

STEP 1A. Check for an active fault code.

Conditions :

- Turn keyswitch ON
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
Check for an active fault code. • Use INSITE™ electronic service tool to read the fault codes.	Fault Code 2185 active? YES	2A
	Fault Code 2185 active? NO	Use the following procedure for an inactive or intermittent fault code. Refer to Procedure 019-362 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-362.html)

STEP 2. Check the ECM and engine harness.

STEP 2A. Inspect the ECM and engine harness connector pins.

Conditions :

- Turn keyswitch OFF.
- Disconnect the engine harness connector from the ECM 60-pin connector.

Action	Specification/Repair	Next Step

Inspect the engine harness and sensor connector pins for the following: • Loose connector • Corroded pins • Bent or broken pins • Pushed back or expanded pins • Moisture in or on the connector • Missing or damaged connector seals • Dirt or debris in or on the connector pins • Connector shell broken • Wire insulation damage • Damaged connector locking tab. Use the following procedure for general inspection techniques. Refer to Procedure 019-361 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-361.html)	Dirty or damaged pins? YES Repair : A damaged connection has been detected in the ECM connector or the engine harness. Clean the connector and pins. Refer to the circuit diagram or wiring diagram for all harness interconnections. • Replace the damaged section of the engine harness. Refer to Procedure 019-043 in Section 19. (/qs3/pubsys2/xml/en/procedures/123/123-019-043.html) • Replace the ECM. Refer to Procedure 019-031 in Section 19. (/qs3/pubsys2/xml/en/procedures/105/105-019-031.html)	3A
	Dirty or damaged pins? NO	2B

STEP 2B. Check the ECM response.

Conditions : • Turn keyswitch OFF • Disconnect the engine harness from the ECM 60-pin connector. • Turn keyswitch ON. • Connect INSITE™ electronic service tool.		
Action	Specification/Repair	Next Step

Check for the appropriate ECM response after 30 seconds. • Use INSITE™ electronic service tool to read the fault codes.	Fault Code 2185 active? YES Repair : Replace the ECM. Refer to Procedure 019-031 in Section 19. (/qs3/pubsys2/xml/en/procedures/105/105-019-031.html)	3A
	Fault Code 2185 active? NO Repair : Replace the damaged section of the engine harness. Refer to Procedure 019-043 in Section 19. (/qs3/pubsys2/xml/en/procedures/123/123-019-043.html) Refer to the circuit diagram or wiring diagram for all harness interconnections.	3A

STEP 3. Clear the fault codes.

STEP 3A. Disable the fault code.

Conditions :

- Connect all components
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
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Disable the fault code. Use INSITE™ electronic service tool to verify the fault code is inactive.	Fault Code 2185 inactive? YES	3B
	Fault Code 2185 inactive? NO Repair : Return to the troubleshooting steps or contact a Cummins® Authorized Repair Location if all steps have been completed and checked again.	1A

STEP 3B. Clear the inactive fault codes.

Conditions :

- Connect all components
- Turn keyswitch ON
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
Clear the inactive fault codes. Use INSITE™ electronic service tool to clear the inactive fault codes.	All fault codes cleared? YES	Repair complete
	All fault codes cleared? NO Repair : Troubleshoot any remaining active fault codes.	Appropriate troubleshooting steps